CLUSTERING INTERVIEW QUESTIONS WITH ANSWERS & EXPLANATIONS

1. Implement K-Means from Scratch

Explanation: Tests understanding of centroid initialization, distance measurement, and iterative optimization.

Answer: K-Means works by initializing K centroids, assigning each point to the nearest centroid, updating centroids as the mean of assigned points, and repeating until convergence.

2. Find Optimal K (Elbow Method)

Explanation: Shows if candidate can evaluate cluster quality.

Answer: Compute inertia (WCSS) for K=1–10 and choose the elbow point where reduction in inertia slows down.

3. Silhouette Score Calculation

Explanation: Checks understanding of separation and cohesion.

Answer: Silhouette score measures how close points are within a cluster vs other clusters. Higher = better clustering.

4. Perform Agglomerative Clustering + Plot Dendrogram

Explanation: Tests hierarchical clustering theory + practice.

Answer: Use scipy.linkage and dendrogram; interpret largest vertical gap to choose cluster count.

5. Compare KMeans vs Agglomerative

Explanation: Tests conceptual clarity.

Answer: KMeans is centroid-based and scalable; agglomerative is hierarchical but computationally expensive.

6. Cluster Customers (Mini Project)

Explanation: Real-world business problem.

Answer: Use KMeans; find optimal K; interpret customer segments (e.g., high income–high spending).

7. DBSCAN Implementation

Explanation: Tests density-based clustering knowledge.

Answer: DBSCAN uses eps and min_samples; identifies core, border, and noise points.

8. Handle Non-Spherical Clusters

Explanation: Tests algorithm selection.

Answer: KMeans fails on non-spherical clusters → use DBSCAN or Hierarchical Clustering.

9. Distance Matrix Creation Manually

Explanation: Checks distance formula knowledge.

Answer: Use Euclidean distance for every pair to build matrix.

10. Predict Cluster for a New Point

Explanation: Tests inference understanding.

Answer: For KMeans, assign new point to nearest centroid using distance.

11. Feature Scaling Before Clustering

Explanation: Evaluates preprocessing knowledge.

Answer: Scaling avoids domination by large-value features; use StandardScaler.

12. Outlier Detection using Clustering

Explanation: Practical clustering usage.

Answer: DBSCAN labels noise as -1 → treat as outliers.

13. PCA + Clustering

Explanation: Checks dimensionality reduction understanding.

Answer: Apply PCA to reduce noise and improve clustering performance.

14. Cluster Evaluation Without Labels

Explanation: Tests metric knowledge.

Answer: Use silhouette score, Davies-Bouldin Index.

15. Time Complexity Comparison

Explanation: Theoretical understanding.

Answer: KMeans ~ O(n), Agglomerative ~ O(n^2) due to distance matrix updates.